

Rajiv Gandhi University of Health Sciences, Karnataka

Fourth Semester B. Pharm Degree Examination – 16/Oct/2025

Time: Three Hours

Max. Marks: 75

MEDICINAL CHEMISTRY - I

Q.P. CODE: 5014

Your answers should be specific to the questions asked

Draw neat labeled diagrams wherever necessary

All the Questions are compulsory

LONG ESSAYS

2 x 10 = 20 Marks

1. Define and classify adrenergic agents with suitable examples and write the synthesis of Salbutamol and Phenylephrine
OR
Define and classify Sedative and Hypnotics with suitable examples and explain the SAR of barbiturates and outline the synthesis of barbital
2. Define and classify NSAIDs with suitable examples and outline the synthesis of Mefenamic acid and Ibuprofen

SHORT ESSAYS

7 x 5 = 35 Marks

3. Explain the role of ionization in relation to biological action
OR
What is Phase I biotransformation? Discuss any two oxidative reactions
4. Discuss hepatic and extra hepatic metabolism
OR
Write the mechanism of action, side effects and uses of Atenolol and Ephedrine
5. Write the biosynthesis and catabolism of acetylcholine
6. Discuss the SAR of parasympathomimetic agents
7. What are cholinergic blocking agents and write the synthesis of Dicyclomine and Procyclidine
8. Define and classify anticonvulsants with suitable examples
9. Discuss the nuclear modification of morphine

SHORT ANSWERS

10 x 2 = 20 Marks

10. Write the aim and purpose of drug metabolism
11. Write the diagrammatic presentation of Cytochrome P450
12. Write the structure of any two alpha adrenergic blockers
13. Write a note on beta receptors
14. **Write the synthesis of Methohexital sodium**
15. **Write the synthesis of Methadone HCl**
16. Write the structure and side effects of Physostigmine
17. Write the mechanism of action of cholinesterase inhibitors
18. Write the structure and uses of Chlorpromazine HCl
19. Write the synthesis of Halothane
